12.3 Conditional Probability; Independent Events; Bayes' Theorem

APPLY IT **What is the probability that an injury sustained by a player while wearing a full shield was a head and face injury?**

By redefining the sample space, we will determine this probability in Example 1.

EXAMPLE 1 Ice Hockey Injuries

The numbers of head and neck injuries for 319 ice hockey players wearing either a full shield or a half shield in the Canadian Inter-University Athletics Union are listed below. (See Exercise 78 in Section 12.1.) ***Source: JAMA.***

Ice Hockey Head and Neck Injuries			
	Half Shield (*H*)	**Full Shield (*F*)**	**Totals**
Head and Face Injuries (A)	95	34	129
Concussions (B)	41	38	79
Neck Injuries (C)	9	7	16
Other Injuries (D)	202	150	352
Totals	347	229	576

(a) If an injured player is selected at random, find the probability that he had a head and face injury as well as the probability that he was wearing a full shield.

SOLUTION Using the letters above, the probability that an injured player had a head and face injury is

$$P(A) = \frac{129}{576} \approx 0.224$$

and the probability that an injured player was wearing a full shield is

$$P(F) = \frac{229}{576} \approx 0.398.$$

(b) Find the probability that an injury sustained by a player while wearing a full shield was a head and face injury.

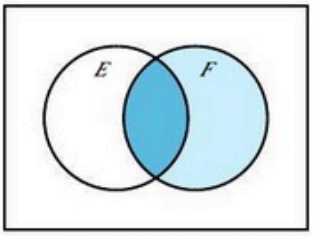
APPLY IT

SOLUTION From the table, of the 229 injuries sustained in players who wore a full shield, 34 injuries were to the head and face, with

$$P(\text{head and face injury while wearing full shield}) = \frac{34}{229} \approx 0.148.$$

The answer to Example 1(b) is a different number from the probability of a head and face injury found in Example 1(a), 0.224, since we have additional information (the player wore a full shield) that has *reduced the sample space*. In other words, we found the probability that an injured player has sustained a head and face injury, A, given that the player was wearing a full shield, F. This is called the *conditional probability* of event A, given that event F has occurred, written $P(A|F)$. The probability $P(A|F)$ is read as "the probability of A given F." In the example above,

$$P(A|F) = \frac{34}{229}.$$



Event F has a total of $n(F)$ elements.

FIGURE 19

To generalize this result, assume that E and F are two events for a particular experiment and that all events in the sample space S are equally likely. We want to find $P(E|F)$, the probability that E occurs given F has occurred. Since we assume that F has occurred, reduce the sample space to F: Look only at the elements inside F. See Figure 19. Of those $n(F)$ elements, there are $n(E \cap F)$ elements where E also occurs. This makes

$$P(E|F) = \frac{n(E \cap F)}{n(F)}.$$

This equation can also be written as the quotient of two probabilities. Divide numerator and denominator by $n(S)$ to get

$$P(E|F) = \frac{n(E \cap F)/n(S)}{n(F)/n(S)} = \frac{P(E \cap F)}{P(F)}.$$

This last result motivates the definition of conditional probability.

Conditional Probability

The **conditional probability** of event E given event F, written $P(E|F)$, is

$$\mathbf{P(E|F) = \frac{P(E \cap F)}{P(F)}, \quad \text{where } P(F) \neq 0.}$$

Although the definition of conditional probability was motivated by an example with equally likely outcomes, it is valid in all cases. However, for *equally likely outcomes*, conditional probability can be found by directly applying the definition, or by first reducing the sample space to event F, and then finding the number of outcomes in F that are also in event E. Thus,

$$P(E|F) = \frac{n(E \cap F)}{n(F)}.$$

In Example 1(b), the conditional probability could have also been found using the definition of conditional probability:

$$P(A|F) = \frac{P(A \cap F)}{P(F)} = \frac{34/576}{229/576} = \frac{34}{229} \approx 0.148.$$

EXAMPLE 2 Hockey

Use the information given in the chart in Example 1 to find the following probabilities.

(a) $P(F|A)$

SOLUTION This represents the probability that an injured player was wearing a full shield given that a head and face injury occurred. Reduce the sample space to A. Then find $n(A \cap F)$ and $n(A)$.

$$P(F|A) = \frac{n(A \cap F)}{n(A)} = \frac{34}{129} \approx 0.264$$

If a head and face injury occurred, then the probability is about 0.264 that the player wore a full shield. This suggests that fewer head and face injuries occur in players who wear a full shield.

(b) $P(A'|F)$

SOLUTION In words, this is the probability that an injury will not be to the head and face for players who wear full shields.

$$P(A'|F) = \frac{n(A' \cap F)}{n(F)} = \frac{195}{229} \approx 0.852$$

(c) $P(F'|A')$

SOLUTION Here, we want the probability that an injured player was not wearing a full shield even though the injury was not to the head and face.

$$P(F'|A') = \frac{n(A' \cap F')}{n(A')} = \frac{252}{447} \approx 0.564$$

TRY YOUR TURN 1

YOUR TURN 1 In Example 2, find $P(A|F')$.

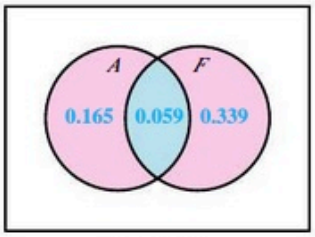


FIGURE 20

Venn diagrams are useful for illustrating problems in conditional probability. A Venn diagram for Example 2, in which the probabilities are used to indicate the number in the set defined by each region, is shown in Figure 20. In the diagram, $P(F|A)$ is found by reducing the sample space to just set A. Then $P(F|A)$ is the ratio of the number in that part of set A which is also in F to the number in set A, or $0.059/0.224 \approx 0.263$. The difference between this value and the one calculated in Example 2(a) is due to rounding.

EXAMPLE 3 Conditional Probabilities

Given $P(E) = 0.4$, $P(F) = 0.5$, and $P(E \cup F) = 0.7$, find $P(E|F)$.

SOLUTION Find $P(E \cap F)$ first. By the union rule,

$$\begin{aligned} P(E \cup F) &= P(E) + P(F) - P(E \cap F) \\ 0.7 &= 0.4 + 0.5 - P(E \cap F) \\ P(E \cap F) &= 0.2. \end{aligned}$$

$P(E|F)$ is the ratio of the probability of that part of E that is in F to the probability of F, or

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{0.2}{0.5} = \frac{2}{5}.$$

TRY YOUR TURN 2

YOUR TURN 2 Given $P(E) = 0.56$, $P(F) = 0.64$, and $P(E \cup F) = 0.80$, find $P(E|F)$.

The Venn diagram in Figure 21 illustrates Example 3.

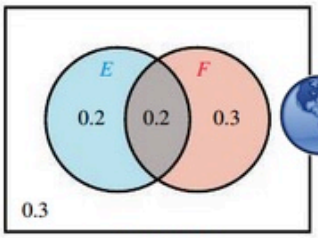


FIGURE 21

EXAMPLE 4 Tossing Coins

Two fair coins were tossed, and it is known that at least one was a head. Find the probability that both were heads.

SOLUTION At first glance, the answer may appear to be 1/2, but this is not the case. The sample space has four equally likely outcomes, $S = \{hh, ht, th, tt\}$. Because of the condition that at least one coin was a head, the sample space is reduced to $\{hh, ht, th\}$. Since only one outcome in this reduced sample space is 2 heads,

$$P(2 \text{ heads} \,|\, \text{at least 1 head}) = \frac{1}{3}.$$

Alternatively, we could use the conditional probability definition. Define two events:

$$E_1 = \text{at least 1 head} = \{hh, ht, th\}$$

and

$$E_2 = 2 \text{ heads} = \{hh\}.$$

YOUR TURN 3 In Example 4, find the probability of exactly one head, given that there is at least one tail.

Since there are four equally likely outcomes, $P(E_1) = 3/4$ and $P(E_1 \cap E_2) = 1/4$. Therefore,

$$P(E_2|E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

TRY YOUR TURN 3

It is important not to confuse $P(A|B)$ with $P(B|A)$. For example, in a criminal trial, a prosecutor may point out to the jury that the probability of the defendant's DNA profile matching that of a sample taken at the scene of the crime, given that the defendant is innocent, $P(D|I)$, is very small. What the jury must decide, however, is the probability that the defendant is innocent, given that the defendant's DNA profile matches the sample, $P(I|D)$. Confusing the two is an error sometimes called "the prosecutor's fallacy," and the 1990 conviction of a rape suspect in England was overturned by a panel of judges, who ordered a retrial, because the fallacy made the original trial unfair. ***Source: New Scientist.***

Later in this section, we will see how to compute $P(A|B)$ when we know $P(B|A)$.

Product Rule

If $P(E) \neq 0$ and $P(F) \neq 0$, then the definition of conditional probability shows that

$$P(E|F) = \frac{P(E \cap F)}{P(F)} \quad \text{and} \quad P(F|E) = \frac{P(F \cap E)}{P(E)}.$$

Using the fact that $P(E \cap F) = P(F \cap E)$, and solving each of these equations for $P(E \cap F)$, we obtain the following rule.

Product Rule of Probability

If E and F are events, then $P(E \cap F)$ may be found by either of these formulas.

$$P(E \cap F) = P(F) \cdot P(E|F) \quad \text{or} \quad P(E \cap F) = P(E) \cdot P(F|E)$$

The product rule gives a method for finding the probability that events E and F both occur, as illustrated by the next few examples.

EXAMPLE 5 Biology Majors

In a class with 2/5 women and 3/5 men, 25% of the women are biology majors. Find the probability that a student chosen from the class at random is a female biology major.

YOUR TURN 4 At a local college, 4/5 of the students live on campus. Of those who live on campus, 25% have cars on campus. Find the probability that a student lives on campus and has a car.

SOLUTION Let B and W represent the events "biology major" and "woman," respectively. We want to find $P(B \cap W)$. By the product rule,

$$P(B \cap W) = P(W) \cdot P(B|W).$$

Using the given information, $P(W) = 2/5 = 0.4$ and $P(B|W) = 0.25$. Thus,

$$P(B \cap W) = 0.4(0.25) = 0.10.$$

TRY YOUR TURN 4

The next examples show how a tree diagram is used with the product rule to find the probability of a sequence of events.

EXAMPLE 6 Environmental Inspections

The Environmental Protection Agency is considering inspecting 6 plants for environmental compliance: 3 in Chicago, 2 in Los Angeles, and 1 in New York. Due to a lack of inspectors, they decide to inspect two plants selected at random, one this month and one next month, with each plant equally likely to be selected, but no plant selected twice. What is the probability that 1 Chicago plant and 1 Los Angeles plant are selected?

SOLUTION A tree diagram showing the various possible outcomes is given in Figure 22. In this diagram, the events of inspecting a plant in Chicago, Los Angeles, and New York are represented by C, LA, and NY, respectively. For the first inspection, $P(\text{C first}) = 3/6 = 1/2$ because 3 of the 6 plants are in Chicago, and all plants are equally likely to be selected. Likewise, $P(\text{LA first}) = 1/3$ and $P(\text{NY first}) = 1/6$.

For the second inspection, we first note that one plant has been inspected and, therefore, removed from the list, leaving 5 plants. For example, $P(\text{LA second}|\text{C first}) = 2/5$, since 2 of the 5 remaining plants are in Los Angeles. The remaining second inspection probabilities are calculated in the same manner.

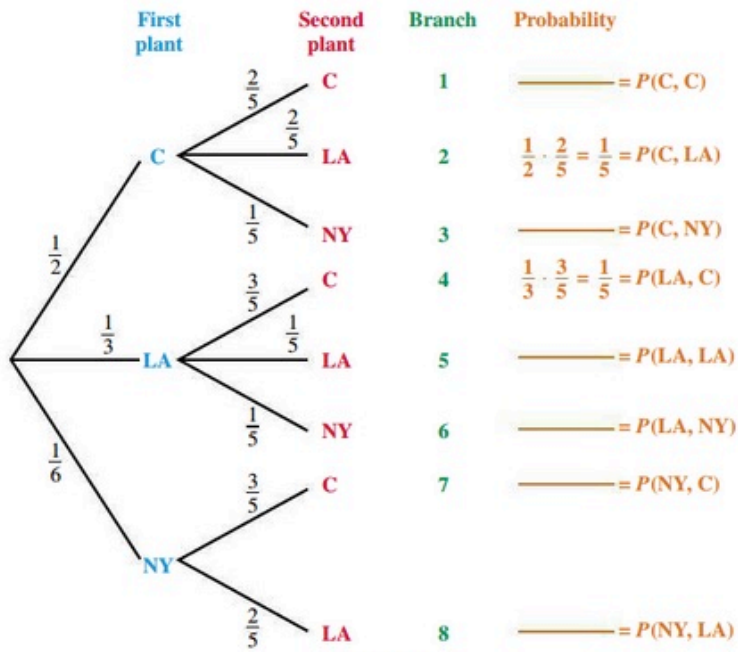


FIGURE 22

We want to find the probability of selecting exactly 1 Chicago plant and 1 Los Angeles plant. This event can occur in two ways: inspecting Chicago this month and Los Angeles next month (branch 2 of the tree diagram), or inspecting Los Angeles this month and Chicago next month (branch 4). For branch 2,

$$P(\text{C first}) \cdot P(\text{LA second}|\text{C first}) = \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{5}.$$

For branch 4, where Los Angeles is inspected first,

$$P(\text{LA first}) \cdot P(\text{C second}|\text{LA first}) = \frac{1}{3} \cdot \frac{3}{5} = \frac{1}{5}.$$

Since the two events are mutually exclusive, the final probability is the sum of these two probabilities.

$$\begin{aligned} P(1\text{ C}, 1\text{ LA}) &= P(\text{C first}) \cdot P(\text{LA second}|\text{C first}) \\ &\quad + P(\text{LA first}) \cdot P(\text{C second}|\text{LA first}) \\ &= \frac{2}{5} \end{aligned}$$

YOUR TURN 5 In Example 6, what is the probability that 1 New York plant and 1 Chicago plant are selected?

TRY YOUR TURN 5

FOR REVIEW

You may wish to refer to the picture of a deck of cards shown in Figure 16 (Section 12.2) and the description accompanying it.

The product rule is often used with *stochastic processes*, which are mathematical models that evolve over time in a probabilistic manner. For example, selecting factories at random for inspection is such a process, in which the probabilities change with each successive selection.

EXAMPLE 7 Playing Cards

Two cards are drawn from a standard deck, one after another without replacement.

(a) Find the probability that the first card is a heart and the second card is red.

SOLUTION Start with the tree diagram in Figure 23. On the first draw, since there are 13 hearts among the 52 cards, the probability of drawing a heart is $13/52 = 1/4$. On the second draw, since a (red) heart has been drawn already, there are 25 red cards in the remaining 51 cards. Thus, the probability of drawing a red card on the second draw, given that the first is a heart, is $25/51$. By the product rule of probability,

$$\begin{aligned} &P(\text{heart first and red second}) \\ &\quad = P(\text{heart first}) \cdot P(\text{red second} \mid \text{heart first}) \\ &\quad = \frac{1}{4} \cdot \frac{25}{51} = \frac{25}{204} \approx 0.123. \end{aligned}$$

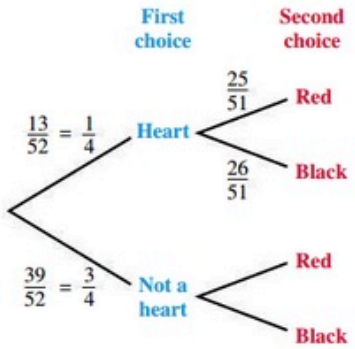


FIGURE 23

(b) Find the probability that the second card is red.

SOLUTION To solve this, we need to fill out the bottom branch of the tree diagram in Figure 23. Unfortunately, if the first card is not a heart, it is not clear how to find the probability that the second card is red, because it depends upon whether the first card is red or black. One way to solve this problem would be to divide the bottom branch into two separate branches: diamond and black card (club or spade).

There is a simpler way, however, since we don't care whether or not the first card is a heart, as we did in part (a). Instead, we'll consider whether the first card is red or black and then do the same for the second card. The result, with the corresponding probabilities, is in Figure 24. The probability that the second card is red is found by multiplying the probabilities along the two branches and adding.

$$\begin{aligned} P(\text{red second}) &= \frac{1}{2} \cdot \frac{25}{51} + \frac{1}{2} \cdot \frac{26}{51} \\ &= \frac{1}{2} \end{aligned}$$

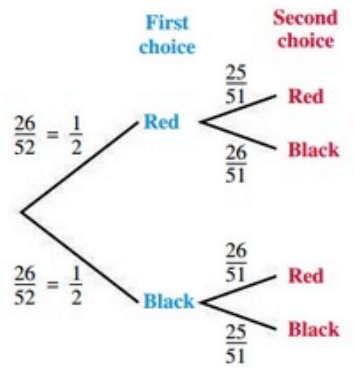


FIGURE 24

The probability is 1/2, exactly the same as the probability that any card is red. If we know nothing about the first card, there is no reason for the probability of the second card to be anything other than 1/2.

Independent Events

Suppose, in Example 7(a), that we draw the two cards *with* replacement rather than without replacement (that is, we put the first card back before drawing the second card). If the first card is a heart, then the probability of drawing a red card on the second draw is $26/52$, rather than $25/51$, because there are still 52 cards in the deck, 26 of them red. In this case, $P(\text{red second} \mid \text{heart first})$ is the same as $P(\text{red second})$. The value of the second card is not affected by the value of the first card. We say that the event that the second card is red is *independent* of the event that the first card is a heart, since the knowledge of the first card does not influence what happens to the second card. On the other hand, when we draw without replacement, the events that the first card is a heart and that the second card is red are *dependent* events. The fact that the first card is a heart means there is one less red card in the deck, influencing the probability that the second card is red.

As another example, consider tossing a fair coin twice. If the first toss shows heads, the probability that the next toss is heads is still 1/2. Coin tosses are independent events, since the outcome of one toss does not influence the outcome of the next toss. Similarly, rolls of a fair die are independent events.

On the other hand, the events "the milk is old" and "the milk is sour" are dependent events; if the milk is old, there is an increased chance that it is sour. Also, in Example 1, the events A (player sustained a head or face injury) and F (player wore a full shield) are dependent events, because information about the use of shields affected the probability of a head or face injury. That is, $P(A|F)$ is different from $P(A)$.

If events E and F are independent, then the knowledge that E has occurred gives no (probability) information about the occurrence or nonoccurrence of event F. That is, $P(F)$ is exactly the same as $P(F|E)$, or

$$P(F|E) = P(F).$$

This, in fact, is the formal definition of independent events.

Independent Events

Events E and F are **independent events** if

$$P(F|E) = P(F) \quad \text{or} \quad P(E|F) = P(E).$$

If the events are not independent, they are **dependent events**.

When E and F are independent events, then $P(F|E) = P(F)$ and the product rule becomes

$$P(E \cap F) = P(E) \cdot P(F|E) = P(E) \cdot P(F).$$

Conversely, if this equation holds, then it follows that $P(F) = P(F|E)$. Consequently, we have this useful fact:

Product Rule for Independent Events

Events E and F are independent events if and only if

$$P(E \cap F) = P(E) \cdot P(F).$$

EXAMPLE 8 Product Rule

For a particular couple expecting their first child, the probability the child will have blue eyes is 0.25 and the probability the child will be a boy is 0.5. Find the probability that the child will be a blue-eyed boy.

SOLUTION If the child having blue eyes and the child being a boy are independent events, then

$$\begin{aligned} P(\text{blue-eyed boy}) &= P(\text{child has blue eyes}) \cdot P(\text{child is a boy}) \\ &= (0.25)(0.5) = 0.125. \end{aligned}$$

TRY YOUR TURN 6

YOUR TURN 6 The probability that you roll a five on a single die is 1/6. Find the probability you roll two fives in a row.

CAUTION It is common for students to confuse the ideas of *mutually exclusive* events and *independent* events. Events E and F are mutually exclusive if $E \cap F = \emptyset$. For example, if a family has exactly one child, the only possible outcomes are $B = \{\text{boy}\}$ and $G = \{\text{girl}\}$. These two events are mutually exclusive. The events are *not* independent, however, since $P(G|B) = 0$ (if a family with only one child has a boy, the probability it has a girl is then 0). Since $P(G|B) \neq P(G)$, the events are not independent.

Of all the families with exactly two children, the events $G_1 = \{\text{first child is a girl}\}$ and $G_2 = \{\text{second child is a girl}\}$ are independent, since $P(G_2|G_1)$ equals $P(G_2)$. However, G_1 and G_2 are not mutually exclusive, since $G_1 \cap G_2 = \{\text{both children are girls}\} \neq \emptyset$.

To show that two events E and F are independent, show that $P(F|E) = P(F)$ or that $P(E|F) = P(E)$ or that $P(E \cap F) = P(E) \cdot P(F)$. Another way is to observe that knowledge of one outcome does not influence the probability of the other outcome, as we did for coin tosses.

NOTE In some cases, it may not be apparent from the physical description of the problem whether two events are independent or not. For example, it is not obvious whether the event that a baseball player gets a hit tomorrow is independent of the event that he got a hit today. In such cases, it is necessary to calculate whether $P(F|E) = P(F)$, or, equivalently, whether $P(E \cap F) = P(E) \cdot P(F)$.

EXAMPLE 9 Snow in Manhattan

On a typical January day in Manhattan the probability of snow is 0.10, the probability of a traffic jam is 0.80, and the probability of snow or a traffic jam (or both) is 0.82. Are the event "it snows" and the event "a traffic jam occurs" independent?

YOUR TURN 7 The probability that you do your math homework is 0.8, the probability that you do your history assignment is 0.7, and the probability of you doing your math homework or your history assignment is 0.9. Are the events "do your math homework" and "do your history assignment" independent?

SOLUTION Let S represent the event "it snows" and T represent the event "a traffic jam occurs." We must determine whether

$$P(T|S) = P(T) \qquad \text{or} \qquad P(S|T) = P(S).$$

We know $P(S) = 0.10$, $P(T) = 0.8$, and $P(S \cup T) = 0.82$. We can use the union rule (or a Venn diagram) to find $P(S \cap T) = 0.08$, $P(T|S) = 0.8$, and $P(S|T) = 0.1$. Since

$$P(T|S) = P(T) = 0.8 \qquad \text{and} \qquad P(S|T) = P(S) = 0.1,$$

the events "it snows" and "a traffic jam occurs" are independent. TRY YOUR TURN 7

Although we showed $P(T|S) = P(T)$ and $P(S|T) = P(S)$ in Example 9, only one of these results is needed to establish independence. It is also important to note that independence of events does not necessarily follow intuition; it is established from the mathematical definition of independence.

Bayes' Theorem

Suppose the probability that a person gets lung cancer, given that the person smokes a pack or more of cigarettes daily, is known. For a research project, it might be necessary to know the probability that a person smokes a pack or more of cigarettes daily, given that the person has lung cancer. More generally, if $P(E|F)$ is known for two events E and F, then $P(F|E)$ can be found using a tree diagram. Since $P(E|F)$ is known, the first outcome is either F or F'. Then for each of these outcomes, either E or E' occurs, as shown in Figure 25.

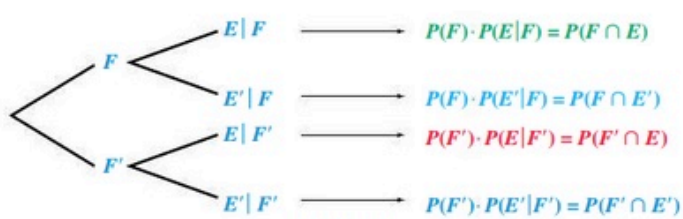


FIGURE 25

The four cases have the probabilities shown on the right. Notice $P(E \cap F)$ is the first case and $P(E)$ is the sum of the first and third cases in the tree diagram. By the definition of conditional probability,

$$P(F|E) = \frac{P(F \cap E)}{P(E)} = \frac{P(F) \cdot P(E|F)}{P(F) \cdot P(E|F) + P(F') \cdot P(E|F')}.$$

This result is a special case of Bayes' theorem, which is generalized later in this section.

Bayes' Theorem (Special Case)

$$P(F|E) = \frac{P(F) \cdot P(E|F)}{P(F) \cdot P(E|F) + P(F') \cdot P(E|F')}$$

EXAMPLE 10 Worker Errors

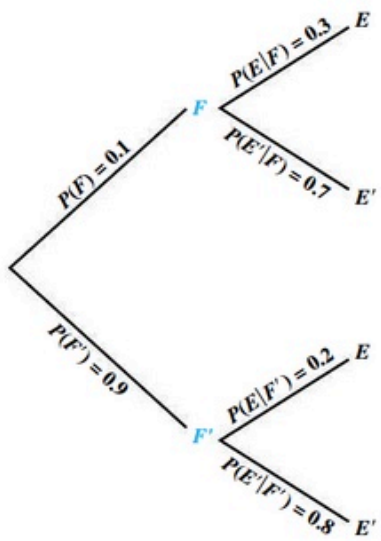
For a fixed length of time, the probability of a worker error on a certain production line is 0.1, the probability that an accident will occur when there is a worker error is 0.3, and the probability that an accident will occur when there is no worker error is 0.2. Find the probability of a worker error if there is an accident.

SOLUTION Let E represent the event of an accident, and let F represent the event of worker error. From the information given,

$$P(F) = 0.1, \qquad P(E|F) = 0.3, \qquad \text{and} \qquad P(E|F') = 0.2.$$

These probabilities are shown on the tree diagram in Figure 26.

YOUR TURN 8 The probability that a student will pass a math exam is 0.8 if he or she attends the review session and 0.65 if he or she does not attend the review session. Sixty percent of the students attend the review session. What is the probability that, given a student passed, the student attended the review session?



Branch	Probability
1	$P(F) \cdot P(E\|F) = P(F \cap E)$
2	$P(F) \cdot P(E'\|F) = P(F \cap E')$
3	$P(F') \cdot P(E\|F') = P(F' \cap E)$
4	$P(F') \cdot P(E'\|F') = P(F' \cap E')$

FIGURE 26

Find $P(F|E)$ by dividing the probability that both E and F occur, given by branch 1, by the probability that E occurs, given by the sum of branches 1 and 3.

$$P(F|E) = \frac{P(F) \cdot P(E|F)}{P(F) \cdot P(E|F) + P(F') \cdot P(E|F')}$$
$$= \frac{(0.1)(0.3)}{(0.1)(0.3) + (0.9)(0.2)} = \frac{0.03}{0.21} = \frac{1}{7} \approx 0.1429$$

TRY YOUR TURN 8

The special case of Bayes' theorem can be generalized to more than two events with the tree diagram in Figure 27. This diagram shows the paths that can produce an event E. We assume that the events $F_1, F_2, \ldots, F_n$ are mutually exclusive events (that is, disjoint events) whose union is the sample space, and that E is an event that has occurred. See Figure 28.

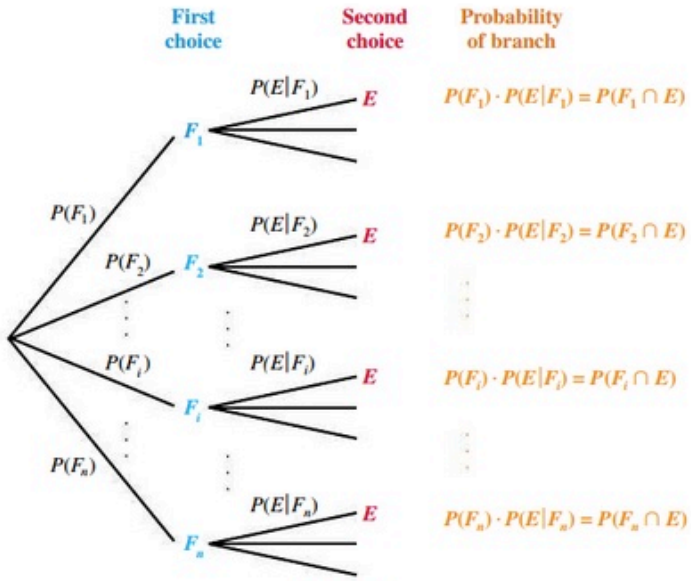


FIGURE 27

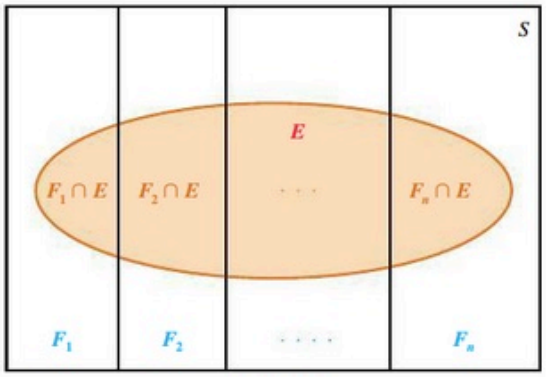


FIGURE 28

The probability $P(F_i|E)$, where $1 \le i \le n$, can be found by dividing the probability for the branch containing $P(E|F_i)$ by the sum of the probabilities of all the branches producing event E.

Bayes' Theorem

$$P(F_i|E) = \frac{P(F_i) \cdot P(E|F_i)}{P(F_1) \cdot P(E|F_1) + P(F_2) \cdot P(E|F_2) + \cdots + P(F_n) \cdot P(E|F_n)}$$

This result is known as **Bayes' theorem**, after the Reverend Thomas Bayes (1702–1761), whose paper on probability was published about three years after his death.

The statement of Bayes' theorem can be daunting. Actually, it is easier to remember the formula by thinking of the tree diagram that produced it. Use the following steps.

Using Bayes' Theorem

1. Start a tree diagram with branches representing $F_1, F_2, \ldots, F_n$. Label each branch with its corresponding probability.
2. From the end of each of these branches, draw a branch for event E. Label this branch with the probability of reaching it, $P(E|F_i)$.
3. You now have n different paths that result in event E. Next to each path, put its probability—the product of the probabilities that the first branch occurs, $P(F_i)$, and that the second branch occurs, $P(E|F_i)$; that is, the product $P(F_i) \cdot P(E|F_i)$, which equals $P(F_i \cap E)$.
4. $P(F_i|E)$ is found by dividing the probability of the branch for F_i by the sum of the probabilities of all the branches producing event E.

Sensitivity and Specificity

The **sensitivity** of a medical test is defined as the probability that a test will be positive given that a person has a disease, written $P(T^+|D^+)$. The **specificity** of a test is defined as the probability that a test will be negative given that

a. Assume that initially each of the prisoners is equally likely to be set free. Assume also that if both B and C are to be executed, the jailer is equally likely to name either B or C. Show that A is wrong, and that his probability of being freed, given that the jailer says B will be executed, is still 1/3.

b. Now assume that initially the probabilities of A, B, and C being freed are 1/4, 1/4, and 1/2, respectively. As in part a, assume also that if both B and C are to be executed, the jailer is equally likely to name either B or C. Now show that A's probability of being freed, given that the jailer says B will be executed, actually drops to 1/5. Discuss the reasonableness of this answer, and why this result might violate someone's intuition.

YOUR TURN ANSWERS

1. 0.274	**2.** 0.625
3. 2/3	**4.** 1/5
5. 1/5	**6.** 1/36
7. No	**8.** 0.6486

12.4 Discrete Random Variables; Applications to Decision Making

APPLY IT **Should patients with malaria-related anemia receive blood transfusions?**
In Example 8, we will calculate the expected values of recovery outcomes and use decision analysis to answer this question.

We shall see that the *expected value* of a probability distribution is a type of average. Probability distributions were introduced briefly earlier in this chapter. Now we take a more complete look at probability distributions. A probability distribution depends on the idea of a *random variable*.

Random Variables

When researchers conduct an experiment, it is necessary to quantify the possible outcomes of the experiment. This process will enable the researcher to recognize individual outcomes and analyze the data. The most common way to record the individual outcomes of the experiment is to assign a numerical value to each of the different possible outcomes of the experiment. For example, if a coin is tossed 2 times, the possible outcomes are: *hh*, *ht*, *th*, and *tt*. For each of these possible outcomes, we could record the number of heads. Then the outcome, which we will label x, is one of the numbers 0, 1, or 2. Of course, we could have used other numbers, like 00, 01, 10, and 11, to indicate these same outcomes, but the values of 0, 1, and 2 are simpler and provide an immediate description of the exact outcome of the experiment. Notice that using this random variable also gives us a way to readily know how many tails occurred in the experiment. Thus, in some sense, the values of x are random, so x is called a **random variable**.

Random Variable

A **random variable** is a function that assigns a real number to each outcome of an experiment.

Random variables are generally categorized as either discrete or continuous. A **discrete random variable** is used in experiments where there is either a finite or countably infinite number of simple events. **Continuous random variables** are used to describe situations where the random variable can be assigned any number in a given interval. We will study continuous random variables in the next chapter.

Probability Distribution A table that lists the possible values of a random variable, together with the corresponding probabilities, is called a **probability distribution**. The sum of the probabilities in a probability distribution must always equal 1. (The sum in some distributions may vary slightly from 1 because of rounding.)

EXAMPLE 1 Hospital Satisfaction

Suppose that 100 patients are asked to rate the services provided by a local hospital. The rating options are: 1 (Outstanding), 2 (Good), 3 (Okay), 4 (Poor), and 5 (Terrible). The results of the survey show that 15 people listed the services as Outstanding, 38 as Good, 25 as Okay, 13 as Poor, and 9 as Terrible. Find the probability distribution for the hospital services rating.

SOLUTION Let x represent the discrete random variable "hospital services rating." The possible values of x are 1, 2, 3, 4, and 5. Since 15 of the 100 patients surveyed rated the hospital as Outstanding, the probability of an Outstanding rating is $P(1) = 15/100$.

Likewise, the probability of a Good rating is $P(2) = 38/100$, the probability of an Okay rating is $P(3) = 25/100$, the probability of a Poor rating is $P(4) = 13/100$, and the probability of a Terrible rating is $P(5) = 9/100$.

The results can be put in a table called a probability distribution.

Probability Distribution of Hospital Survey

x	1	2	3	4	5
$P(x)$	15/100	38/100	25/100	13/100	9/100

Instead of writing the probability distribution as a table, we could write the same information as a set of ordered pairs:

$$\{(1, 15/100), (2, 38/100), (3, 25/100), (4, 13/100), (5, 9/100)\}.$$

There is just one probability for each value of the random variable. Thus, a probability distribution defines a function, called a **probability distribution function**, or simply a **probability function**. We shall use the terms "probability distribution" and "probability function" interchangeably.

The information in a probability distribution is often displayed graphically as a special kind of bar graph called a **histogram**. The bars of a histogram all have the same width, usually 1. (The widths might be different from 1 when the values of the random variable are not consecutive integers.) The heights of the bars are determined by the probabilities. A histogram for the data in Example 1 is given in Figure 31. A histogram shows important characteristics of a distribution that may not be readily apparent in tabular form, such as the relative sizes of the probabilities and any symmetry in the distribution.

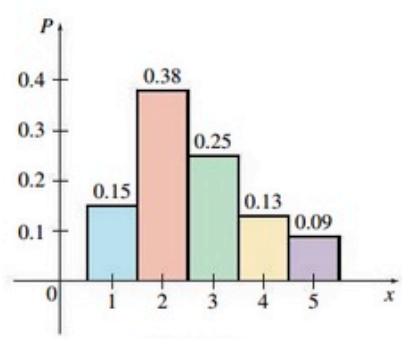


FIGURE 31

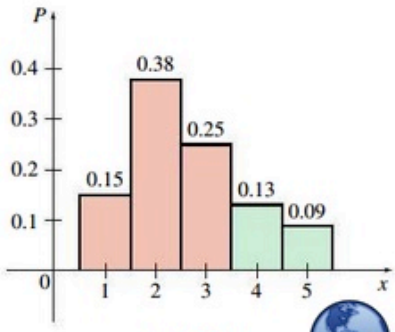


FIGURE 32

The area of the bar above $x = 1$ in Figure 31 is the product of 1 and 15/100, or $1 \cdot 15/100 = 0.15$. Since each bar has a width of 1, its area is equal to the probability that corresponds to that value of x. The probability that a particular value will occur is thus given by the area of the appropriate bar of the graph. For example, the probability that a person rated the hospital as Outstanding, Good, or Okay is the sum of the areas for $x = 1$, $x = 2$, and $x = 3$. This area, shown in pink in Figure 32, corresponds to 78/100, which corresponds to 78% of the total area, since

$$P(x \le 3) = P(x = 1) + P(x = 2) + P(x = 3).$$

EXAMPLE 2 Probability Distributions

(a) Give the probability distribution for the number of heads showing when two coins are tossed.

SOLUTION Let x represent the random variable "number of heads." Then x can take on the values 0, 1, or 2. Now find the probability of each outcome. To find the probability of 0, 1, or 2 heads, notice that there are 4 outcomes in the sample space: $\{hh, ht, th, tt\}$. The results are shown in the table with Figure 33.

Probability Distribution of Heads			
x	0	1	2
$P(x)$	1/4	1/2	1/4

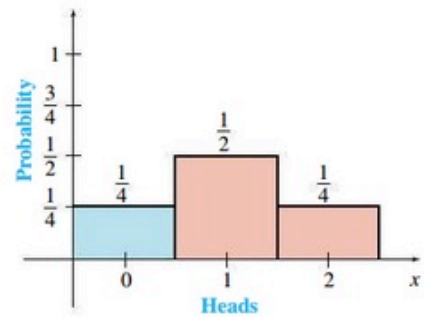


FIGURE 33

(b) Draw a histogram for the distribution in the table. Find the probability that at least one coin comes up heads.

SOLUTION The histogram is shown in Figure 33. The portion in pink represents

$$P(x \ge 1) = P(x = 1) + P(x = 2)$$
$$= \frac{3}{4}.$$

TRY YOUR TURN 1

YOUR TURN 1 Find the probability distribution and draw a histogram for the number of tails showing when three coins are tossed.

Expected Value

In working with probability distributions, it is useful to have a concept of the typical or average value that the random variable takes on. In Example 2, for instance, it seems reasonable that, on the average, one head shows when two coins are tossed. This does not tell what will happen the next time we toss two coins; we may get two heads, or we may get none. If we tossed two coins many times, however, we would expect that, in the long run, we would average about one head for each toss of two coins.

A way to solve such problems in general is to imagine flipping two coins 4 times. Based on the probability distribution in Example 2, we would expect that 1 of the 4 times we would get 0 heads, 2 of the 4 times we would get 1 head, and 1 of the 4 times we would get 2 heads. The total number of heads we would get, then, is

$$0 \cdot 1 + 1 \cdot 2 + 2 \cdot 1 = 4.$$

The expected numbers of heads per toss is found by dividing the total number of heads by the total number of tosses, or

$$\frac{0 \cdot 1 + 1 \cdot 2 + 2 \cdot 1}{4} = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1.$$

Notice that the expected number of heads turns out to be the sum of the three values of the random variable x multiplied by their corresponding probabilities. We can use this idea to define the *expected value* of a random variable as follows. (The expected value of a discrete random variable is also called the mean and is represented by the Greek letter mu, μ.)

Expected Value (or Mean)

Suppose the discrete random variable x can take on the n values $x_1, x_2, x_3, \ldots, x_n$. Also, suppose the probabilities that these values occur are, respectively, $p_1, p_2, p_3, \ldots, p_n$. Then the **expected value** of the random variable is

$$E(x) = \mu = x_1p_1 + x_2p_2 + x_3p_3 + \cdots + x_np_n.$$

In summation notation

$$E(X) = \mu = \sum_x xP(X = x) = \sum_x xP(x),$$

where the sum is taken over all possible values of the random variable X.

EXAMPLE 3 Hospital Satisfaction

In Example 1, find the expected satisfaction rating of the hospital.

SOLUTION Using the values in the table from Example 1 and the definition of expected value, we find that

$$\begin{aligned} E(X) &= 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4) + 5 \cdot P(X = 5) \\ &= 1 \cdot \frac{15}{100} + 2 \cdot \frac{38}{100} + 3 \cdot \frac{25}{100} + 4 \cdot \frac{13}{100} + 5 \cdot \frac{9}{100} = \frac{263}{100} = 2.63. \end{aligned}$$

Thus, the average person will rate the hospital between Good and Okay, but closer to Okay.

Physically, the expected value of a probability distribution represents a balance point. If we think of the histogram in Figure 31 as a series of weights with magnitudes represented by the heights of the bars, then the system would balance if supported at the point corresponding to the expected value.

EXAMPLE 4 Symphony Orchestra

Suppose a local symphony decides to raise money by raffling an HD television worth \$400, a dinner for two worth \$80, and 2 CDs worth \$20 each. A total of 2000 tickets are sold at \$1 each. Find the expected payback for a person who buys one ticket in the raffle.

Method 1 Direct Calculation

SOLUTION

Here the random variable represents the possible amounts of payback, where payback = amount won − cost of ticket. The payback of the person winning the television is \$400 (amount won) − \$1 (cost of ticket) = \$399. The payback for each losing ticket is \$0 − \$1 = −\$1.

The paybacks of the various prizes, as well as their respective probabilities, are shown in the table on the next page. The probability of winning \$19 is 2/2000 because there are 2 prizes worth \$20. We have not reduced the fractions in order to keep all the denominators equal. Because there are 4 winning tickets, there are 1996 losing tickets, so the probability of winning −\$1 is 1996/2000.

Probability Distribution of Prize Winnings				
x	\$399	\$79	\$19	−\$1
$P(x)$	1/2000	1/2000	2/2000	1996/2000

The expected payback for a person buying one ticket is

$$399\left(\frac{1}{2000}\right) + 79\left(\frac{1}{2000}\right) + 19\left(\frac{2}{2000}\right) + (-1)\left(\frac{1996}{2000}\right) = -\frac{1480}{2000}$$
$$= -0.74.$$

On average, a person buying one ticket in the raffle will lose \$0.74, or 74¢.

It is not possible to lose 74¢ in this raffle: either you lose \$1, or you win a prize worth \$400, \$80, or \$20, minus the \$1 you pay to play. But if you bought tickets in many such raffles over a long period of time, you would lose 74¢ per ticket on average. It is important to note that the expected value of a random variable may be a number that can never occur in any one trial of the experiment.

Method 2
Alternate Procedure

An alternative way to compute expected value in this and other examples is to calculate the expected amount won and then subtract the cost of the ticket afterward. The amount won is either \$400 (with probability 1/2000), \$80 (with probability 1/2000), \$20 (with probability 2/2000), or \$0 (with probability 1996/2000). The expected payback for a person buying one ticket is then

$$400\left(\frac{1}{2000}\right) + 80\left(\frac{1}{2000}\right) + 20\left(\frac{2}{2000}\right) + 0\left(\frac{1996}{2000}\right) - 1 = -\frac{1480}{2000}$$
$$= -0.74.$$

TRY YOUR TURN 2

YOUR TURN 2 Suppose that there is a \$5 raffle with prizes worth \$1000, \$500, and \$250. Suppose 1000 tickets are sold and you purchase a ticket. Find your expected payback for this raffle.

EXAMPLE 5 Friendly Wager

Each day Donna and Mary toss a coin to see who buys coffee (\$1.20 a cup). One tosses and the other calls the outcome. If the person who calls the outcome is correct, the other buys the coffee; otherwise the caller pays. Find Donna's expected payback.

SOLUTION Assume that an honest coin is used, that Mary tosses the coin, and that Donna calls the outcome. The possible results and corresponding probabilities are shown below.

Possible Results				
Result of Toss	Heads	Heads	Tails	Tails
Call	Heads	Tails	Heads	Tails
Caller Wins?	Yes	No	No	Yes
Probability	1/4	1/4	1/4	1/4

Donna wins a \$1.20 cup of coffee whenever the results and calls match, and she loses a \$1.20 cup when there is no match. Her expected payback is

$$(1.20)\left(\frac{1}{4}\right) + (-1.20)\left(\frac{1}{4}\right) + (-1.20)\left(\frac{1}{4}\right) + (1.20)\left(\frac{1}{4}\right) = 0.$$

This implies that, over the long run, Donna neither wins nor loses.

A game with an expected value of 0 (such as the one in Example 5) is called a **fair game**. Casinos do not offer fair games. If they did, they would win (on average) \$0, and have a hard time paying the help! Casino games have expected winnings for the house that vary from 1.5 cents per dollar to 60 cents per dollar. Exercises 35–37 at the end of the section ask you to find the expected payback for certain games of chance.

The idea of expected value can be very useful in decision making, as shown by the next example.

EXAMPLE 6 Life Insurance

At age 50, you receive a letter from Mutual of Mauritania Insurance Company. According to the letter, you must tell the company immediately which of the following two options you will choose: take \$20,000 at age 60 (if you are alive, \$0 otherwise) or \$30,000 at age 70 (again, if you are alive, \$0 otherwise). Based *only* on the idea of expected value, which should you choose?*

SOLUTION Life insurance companies have constructed elaborate tables showing the probability of a person living a given number of years into the future. From a recent such table, the probability of living from age 50 to 60 is 0.88, while the probability of living from age 50 to 70 is 0.64. The expected values of the two options are given below.

$$\text{First option: } (20{,}000)(0.88) + (0)(0.12) = 17{,}600$$

$$\text{Second option: } (30{,}000)(0.64) + (0)(0.36) = 19{,}200$$

Based strictly on expected values, choose the second option.

Variance

Although the expected value provides a measure of central tendency, it does not provide a measure of how much the data vary from the expected value. For example, the three sets of numbers in the following table all have an average of 50. But, as you can see, the values in the third distribution vary significantly from the mean.

List	Values	Mean
Set 1	15, 25, 70, 90	50
Set 2	45, 46, 47, 48, 49, 51, 52, 53, 54, 55	50
Set 3	1, 17, 45, 70, 117	50

The **variance** (represented by σ^2 or $\text{Var}(X)$) of a probability distribution is a measure of the *spread* of the values of a probability distribution. For a discrete distribution, the variance is found by taking the expected value of the squares of the differences of the values of the random variable and the mean. Thus,

$$\sigma^2 = \text{Var}(X) = \sum_x (x - \mu)^2 P(x),$$

where the sum is taken over all possible values of the random variable X, and μ is the expected value (or mean) of X.

Think of the variance as the expected value of $(X - \mu)^2$, which measures how far X is from the mean μ. The **standard deviation** (represented by σ) of X is defined as

$$\sigma = \sqrt{\sigma^2}.$$

*In reality, the value of money now versus money in the future should be considered. For more on this idea see Chapter 5, Mathematics of Finance, in *Finite Mathematics*, 10th edition, by M. Lial, R. Greenwell, and N. Ritchey, Boston, MA: Pearson, 2012.